When looking to select a specific communication technology for the sensors that will feed data to your digital twin, you should consider the following factors:

- *Coverage requirements*—the distance between sensors and gateways, and what technology will support that distance.
- *Power constraints*—whether you have access to mains power or if batteries are a better choice, and how long batteries might last.
- *Data requirements*—how much data, and at what rate, will you need to communicate.
- *Infrastructure*—what already exists versus what you will need to deploy.
- *Total cost*—including devices, gateways, and ongoing service fees.

Given that most of the sensors I have chosen communicate over LoRaWAN, I needed to configure a private LoRaWAN network. Appendix A goes into the details of how I have done this, using a consumer gateway (the Dragino LPS8N indoor gateway) and leveraging a managed cloud service (AWS IoT Core for LoRaWAN) to provide a managed network server. Where I have access to mains power, I use WiFi for communications. This creates a hybrid home sensor network (figure 3.6) where most sensors are battery-powered and communicate over LoRaWAN, while mains-powered sensors in WiFi coverage areas communicate directly with the message broker.